COVID-19

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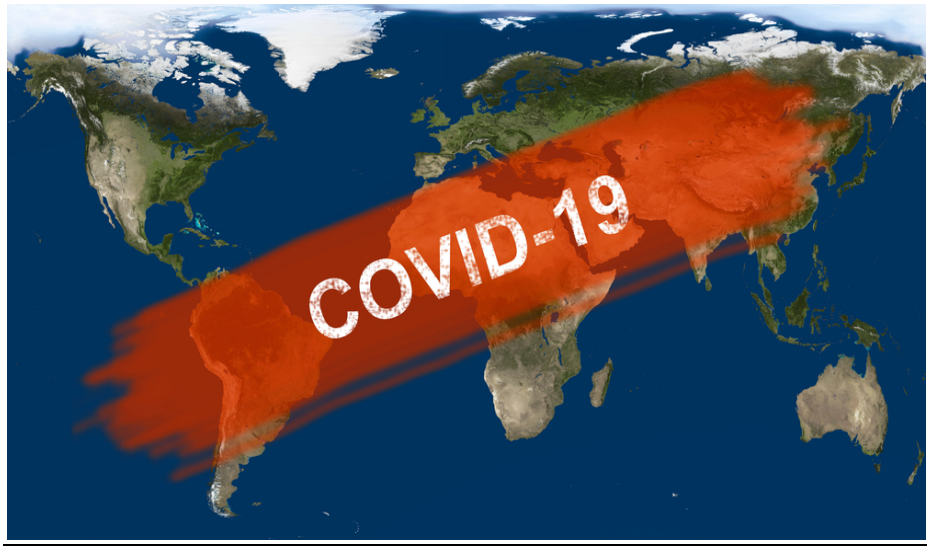

Introduction

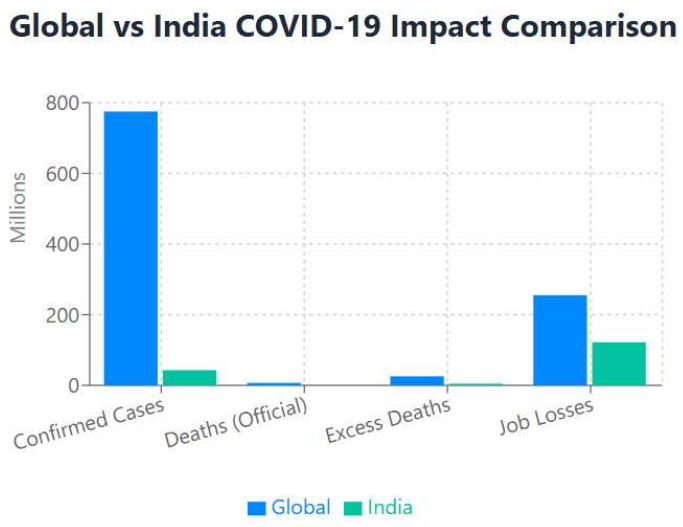

Fig1. No. of cases

The COVID-19 pandemic represents one of the worst global health crisis of the 21st century, altering human civilization across multiple dimensions. Caused by the SARS-CoV-2 virus the first case was found in Wuhan, China in December 2019, the pandemic rapidly evolved from a localized outbreak to a worldwide catastrophe. The World Health Organization firstly declared it as Public Health Emergency of International Concern on January 30, 2020, and then elevated its status to pandemic on 11th March 2020. Till December 2022, over 500 million cases were confirmed globally with approximately 6.6 million reported deaths, though actual mortality figures are estimated to be substantially higher due to undercounting and limited surveillance in many regions of the world.(Wikipedia,2026)

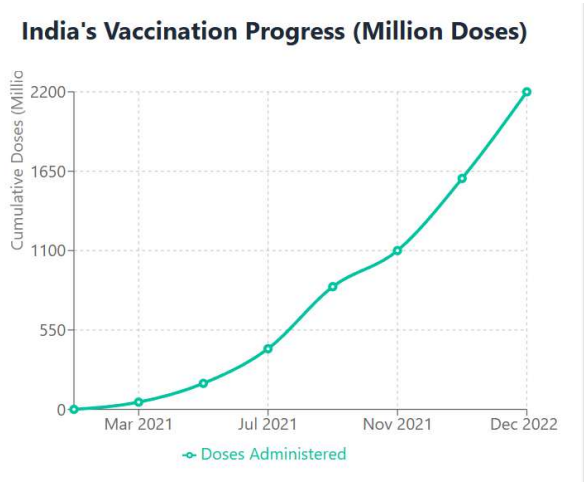

Fig2. Vaccination in India

In India the first case was reported on 30th of January 2020, in Kerala when a student returned to India from Wuhan. The pandemic later spread throughout the country in waves with the second wave of April-May 2021 being the worse due to the rapid spread of Delta variant. In this phase healthcare systems were on the verge of collapse, a nightmare scenario of oxygen shortages, overcrowded crematoriums, and desperate families in search of beds at hospitals became the symbol of the disaster. The Indian reaction involved one of the strictest lockdowns in the world. There was an enormous increase and development of the testing infrastructure. The number of tests increased to 1.5 lakh tests per day and also the largest vaccination drive in human history with more than 220 crore doses were administered.(Wikipedia,2026)

Disaster Typology and Hazard Profile

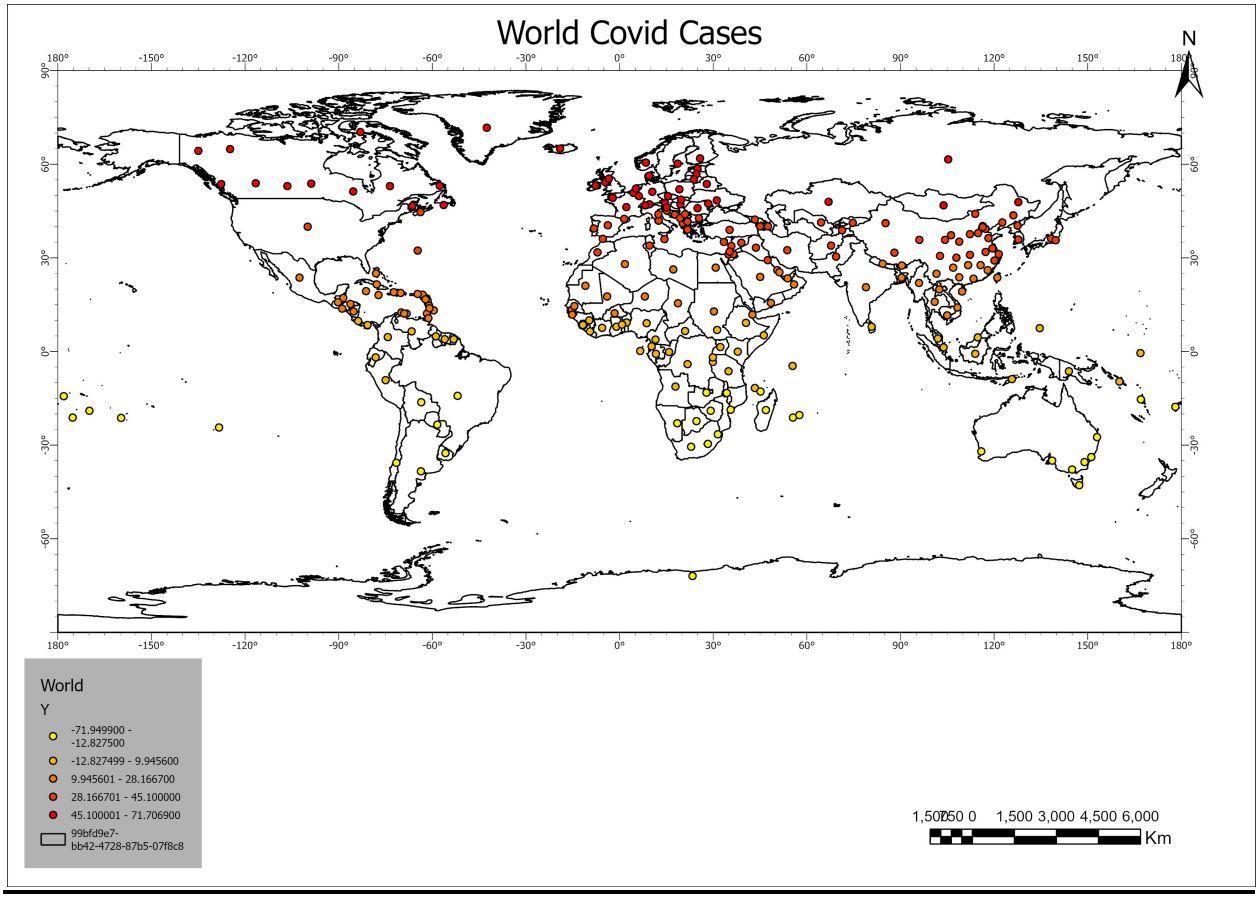

Fig3. Map showing Covid 19 across the world

COVID-19 is marked as a biological disaster, particularly a global pandemic caused by the SARS-CoV-2 virus. According to WHO, a pandemic is defined by sustained worldwide transmission across multiple regions, overwhelming normal response capacities. What makes

COVID-19 special is that it was also a compound disaster meaning as it started as a health problem but triggered many other problems like economic collapse, food shortages, education disruption, and social unrest. The interrelatedness of effects is what makes the difference between pandemics and more local or sector-specific disasters, and requires the inclusion of multisectoral response frameworks. The virus spread mainly through air when person breathes, talks, coughs, or sneezes as small water droplets and aerosols having large life span ranging from hours to days carried the carried the virus from one person to another. This is why wearing masks maintaining distance and disinfecting surfaces became important protective measures. (WHO,2020)

Indicators and Data Sources

Indicator	Unit	Global (2020-22)	India	Source
Confirmed Cases	Millions	775	43M	WHO,2025
Deaths (Official)	Millions	6.9	0.53M	WHO,2025
Excess Deaths	Millions	18-33	4.7M	The Economist, 2021
GDP Loss	$ Trillions	16	0.3T	World Bank,2022
Vaccination doses	Billions	13.5	2.2B	WHO,2025
Job Losses	Millions	255	122M	ILOSTAT,
Internal Displacement	Millions	12	6M migrants	EM-DAT,2022

Table1. Showing data of World and India

These indicators can be used to depict the implications of the COVID-19 pandemic (2020-22) in different sectors such as health, economic and social sectors on the global and Indian levels. The direct health burden is represented by data on confirmed cases and the official number of deaths provided by the World Health Organization, whereas the excess death estimates indicate that there is a high level of underreporting, especially in India. Loss of GDPs provides by world bank shows impact of lockdown on economies. Vaccination statistics indicate the scale of the public health response. The humanitarian outcomes of mobility restrictions can also be further

illustrated through the internal displacement data of EM-DAT which highlights the way COVID-19 was turned into a compound disaster that was much beyond a health emergency.

Causal Analysis (Hazard–Exposure–Vulnerability Framework)

Hazard Characteristics:- Nature of SARS-CoV-2

SARS-CoV-2 was a serious threat as it was a new virus and there a a lack of knowledge about it in thee beginning and since it was new there was no existing immunity the spreadness rate was very. Its high ability to spread (with an R_0(Basic Reproduction number) of about 2-3 for the original strain and 5-8 for the Delta variant) caused the number of infections to increase rapidly.

The situation became worse with the arrival of Delta variant which caused repeated waves of infection in India despite earlier control efforts. Secondly, the virus had long incubation period, and many time no symptoms were seen or it was spread before symptoms appeared making hard to detect and isolate cases early. Although the overall death rate low, but large number of infections meant that even small share of serious cases can put enormous pressure healthcare systems.

Exposure Analysis:- Population, Infrastructure, and Behaviour

The exposure to the virus in India was significantly high because of the population density, and physical distancing was highly challenging particularly in cities and informal settlements. Continuous exposure pathways were brought about by mass transportation, congested markets and crowded housing. Overcrowding, staff shortages and long work hours of frontline workers also made healthcare facilities an exposure point. Most workers were unable to work off site due to informal employment and had a tradeoff between income loss and the risk of infection. Massive movement of people, extended families and mass gatherings such as religious functions and political gatherings were turned into super spreader environments, and they quickly spread the infection to other parts of the globe.

Vulnerability Analysis:- Why Exposure Led to Severe Outcomes

Exposure was transformed into colossal destruction through the vulnerabilities of the health systems. Decades of insufficient investment in the state health resulted in shortcomings in hospital capacity, ICU, oxygen infrastructure, and human resources when the infection rates went high. The second wave oxygen crisis was a revelation of vulnerabilities in the supply chain,

which led to avoidable deaths. Impacts were enhanced by socioeconomic factors. Compliance with preventive measures were challenging as a large informal workforce and overcrowded housing were an obstacle to this. The digital divide has made many of the people unable to obtain online education, telemedicine, welfare, and vaccine registration.Poor governance including sudden lockdowns, weak coordination, and data underreporting combined with high rates of comorbidities like diabetes, TB and malnutrition made COVID-19 both a health crisis and a broader social and economic disaster.

Impacts of COVID-19 in India

Social Impacts

The nationwide lockdown which was announced on 24 March 2020 suddenly brought everyday life to a halt. Migrant workers suffered the most as workplaces shut down and incomes disappeared, leaving them without food, shelter, or support in cities. With public transport halted, millions walked back to their villages, often covering hundreds of kilometers. t is estimated that between 10 and 40 million migrants attempted to go back home, and even the groups such as the Stranded Workers Action Network had received more than 100, 000 distress calls about hunger and uncertainty. (SWAN, 2020).

Learning was greatly disrupted with approximately 320 million learners having to remain at home due to school closures of almost one and a half year. Online classes were encouraged, but about a quarter of households were connected to the internet, which exacerbated the inequalities in education, particularly among rural and poor families (UNESCO, 2021; ASER, 2022).

Economic Impacts

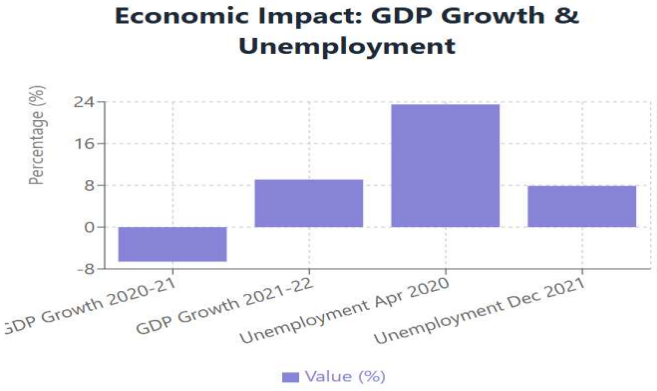

Fig4. GDP and Unemployment in India

Their was a 6.6% contraction in India in 2020-21, the worst drop in the economy since independence. The economic activity was reduced in most sectors during the lockdown. Especially affected were tourism and aviation. In April 2020, the number of unemployed people increased immensely. There was declining demand and saving in small and informal businesses. The daily wage earners and the shop owners could not easily recover and some businesses were closed permanently (Ministry of Statistics and Programme Implementation, 2021; RBI, 2021).

Environmental Impacts

The pandemic in the first few months of its progress appeared to have a positive effect on the environment, as most cities began to breathe cleaner air, and carbon emissions dropped in 2020. These gains were however shortlived. The influx of masks, kits of PPE, and medical garbage became the matter of great concern when it comes to environmental health and safety. (CPCB, 2020; Global Carbon Project, 2021; CPCB, 2021; Down To Earth, 2021)

Psychological and Social Impacts

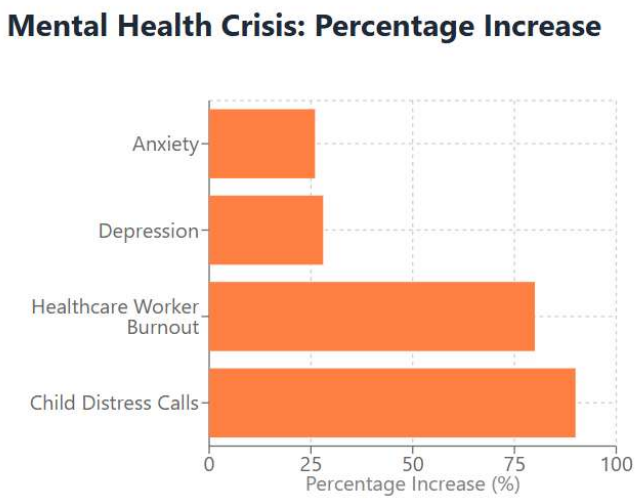

Fig5. Mental Health Crisis

COVID-19 created a mental health crisis that was very prevalent in all populations. Livelihood loss, isolation, and fear of infection resulted in permanent stress. The number of cases of depression rose by 28 percent and anxiety by 26 percent globally and India has an enormous

share of it (The Lancet, 2021). NIMHANS surveys established an almost 40 percent respondents who said they experience anxiety and 15 percent who displayed severe depression. Almost 80 percent of health care personnel have cited burnout at work due to high death rates and workloads.

The stringent lockdown in India lasted 68 days in the initial set, and most of the cities had more than 200 days of restrictions on mobility, which led to chronic stress. The 18 months of school closure disrupted the routine and emotional development of children. Child helplines 90-degree rises in distress calls. Domestic violence as well gained momentum, with the number of complaints to the National Commission of Women going up 2.5 times during lockdown.

The stigma of COVID-19 had a socially harmful impact, and people who contracted the disease, and their families were excluded. Nevertheless, resilience in the community occurred in the form of volunteers who handed out food and oxygen. These changes created a lasting burden as long COVID is seen in 10-30% of those infected with fatigue, cognitive impairment, and breathlessness (Nature, 2022).

Application of the Disaster Management Framework

Mitigation and Preparedness

Pandemics are classified as potential disasters, and prior to COVID-19, India had disaster frameworks under national Disaster Management Act, 2005, and National Disaster management plan (2019). There were disease surveillance systems such as IDSP and NCDC (NDMA, 2020). But preparedness lapses were high: the amount spent on public health was only 1.28 percent of the GDP, the number of hospital beds was approximately 0.5 per 1,000 population and the number of ICU beds was very low. The strength of India was the ability to produce vaccines by the Serum Institute and Bharat Biotech. Massive planning shortfalls were revealed by the nationwide lockdown on 24 March 2020, with just four hours notice (PIB, 2020).

Response

India was implementing one of the most severe lockdowns in the world, which curtailed movement by 70%-80%. Nevertheless, when the sudden shutdown caused a migrant crisis. Relief schemes were used such as the PM Garib Kalyan Yojana that foisted free meals on 80 crore citizens. By 2021, testing capacity had increased to more than 2 million tests a day. Online

applications such as Aarogya Setu and CoWIN aided in contact tracing and immunization. The peak number of cases per day was above 4 lakh in the second wave (April May 2021) and led to oxygen scarcity. Oxygen Express trains were used as emergency measures. By the end of 2022, India had conducted more than 220 crore doses of vaccination (CoWIN Dashboard)

Recovery

GDP recovered to 9.1% growth in 2021-22 with the help of economic recovery (RBI, 2022). Healthcare recovery on fortification of the infrastructure under PM Ayushman Bharat. The National Tele Mental Health Programme boosted mental health services and education recovery focused on learning loss through NIPUN Bharat. There are still challenges, such as disproportionate recovery and long-term effects of Long COVID.

Evaluation of Disaster Management Effectiveness

The response of the disaster management in India to COVID-19 demonstrated an ambiguous picture, as both the response in terms of swift response and magnitude was commendable, whereas the response in terms of planning, inclusion and coordination left much to be desired. Early warning showed a certain level of efficiency: India reported its first case on 30 January 2020, began thermal screening in airports and within weeks the Indian Council of Medical Research (ICMR) created test guidelines. Nonetheless, it was not recognised until March and transmission on a community basis was not officially recognised, testing criteria were very restrictive, and airborne transmission risks were not recognised early, which hampered the effectiveness of early containment.

The restrictions that were implemented throughout the country on 24 March 2020 were effective in curbing the spread of the disease. The daily growth rates went down to approximately 20 percent to about 6-7%, and the doubling period went up, which gave the purchasing time needed to increase the healthcare capacity. There was an augmentation of hospital bed by approximately 35% and an augmentation of ICU bed by 20% and the testing rose to over 150,000 compared to 100 tests per day in June 2020. The sudden lockdown was disastrous in spite of the epidemiological gains. This resulted in the displacement of between 10 and 40 million migrant workers and a reduction in GDP by 24.4% in Q1 of 2020-21, meaning that poor planning has a very high humanitarian cost (The Wire, 2020).

Relief measures were fairly extensive but not equal in sufficiency. The PM Garib Kalyan Anna Yojana was able to touch the hearts of close to 80crore individuals so that there was no mass hunger. But the migrants who did not have ration cards such as the ones estimated to be more than 100 million were mostly left out. The cash transfers were not enough and offered limited support to fulfill the basic needs. MSMEs were supported in terms of collateral free loans, which served formal enterprises, yet informal businesses were not engaged in any significant number as only about 30% of the eligible MSMEs were able to access credit (RBI, 2021).

Inter-governmental coordination was also strong and weak at the same time. The consultations between Centres and States increased coordination in the long run, however, disagreements regarding the procurement of vaccines, distribution of oxygen during the second wave, and uneven reporting of the data weakened trust. Civil-military coordination turned out to be a definite victory, as the assistance of the Indian Railways Oxygen Express and the support of the army was essential.

The results of recovery were partial. Recovery in economic growth had a reverse effect of 9.1% in 202-22, and recovery in the informal sector was still poor. In education, by the end of 2021, schools had reopened, but it was estimated that 11.5 years of learning had been lost in education, more so among rural children and girls (ASER, 2022).

Institutional and Governance Analysis

Government Agencies

The Ministry of Health and Family Welfare (MoHFW) was the main body responsible for making policies and coordinating the COVID-19 response. It provided daily updates, issued health guidelines, worked with international organisations, and managed vaccine approval and distribution. Its strengths included regular public communication, quick guideline updates, and running the national vaccination programme. However, it initially underestimated the pandemic's severity, was slow to acknowledge community transmission and faced criticism over data transparency (MoHFW, 2020; WHO India, 2021).

The National Disaster Management Authority (NDMA) coordinated the overall disaster response under the Disaster Management Act, 2005. It managed resources between different ministries and issued guidelines to state governments. NDMA effectively used existing disaster response

systems during the second wave, particularly through oxygen supply initiatives like Oxygen Express. However, coordination gaps between disaster management and health authorities limited its effectiveness (NDMA, 2020; Ministry of Home Affairs, 2021).

The Indian Council of Medical Research (ICMR) developed testing protocols, expanded laboratory networks from 52 to over 3000, coordinated vaccine trials including Covaxin and provided health guidance. However, strict early testing criteria delay in publishing study findings and controversy over emergency vaccine approval affected public trust (ICMR, 2021).

State responses varied widely. Kerala showed strong early containment through aggressive contact tracing and transparent reporting. Maharashtra faced the highest cases but implemented effective strategies like the Dharavi model. Delhi experienced severe hospital overcrowding and became the centre of the oxygen crisis during the second wave, worsened by Centre-State coordination problems (State Government Reports, 2020–2021).

NGOs and Civil Society Organizations

Over 10,000 NGOs participated in COVID-19 relief across India. They distributed an estimated 1 billion meals, operated community kitchens, supported migrant workers, and provided mental health services. Major organisations included Akshaya Patra Foundation, Goonj, Khalsa Aid, and SEEDS. Mental health helplines by iCALL and Vandrevala Foundation addressed rising psychological distress (CSIP, 2021).

NGOs were effective due to their flexibility, community trust, and ability to reach populations missed by government schemes. However, the lack of formal coordination caused duplication in some areas and gaps in others. Bureaucratic hurdles and uneven funding also limited effectiveness (CSIP, 2021).

International Organizations

International organisations provided critical support. The World Health Organization supplied technical guidance, training, and medical supplies. UNICEF, UNDP, and UNFPA supported oxygen infrastructure, vaccination logistics, and health services. The World Bank provided over USD 2.5 billion for health system strengthening (WHO India, 2021; World Bank, 2021).

During the second wave, 76 countries provided oxygen concentrators, medicines, and equipment. India's shift from vaccine donor under Vaccine Maitri to aid recipient highlighted challenges in global vaccine equity and pandemic preparedness (Ministry of External Affairs, 2021).

Risk Reduction and Control Measures

Structural Measures

Strengthening healthcare infrastructure is essential for reducing future risks. ICU capacity must increase from the current 0.7 beds per 10,000 people to at least 3 beds per 10,000. Every district hospital should have a minimum of 20 ventilator-equipped ICU beds, with modular units ready for rapid deployment during emergencies. The second wave showed the consequences of inadequate critical care—ICU occupancy in cities like Delhi reached 99%, resulting in many patients being denied treatment. This would require an estimated investment of ₹50,000 crore over five years, using public–private partnerships in cities and central assistance for 272 aspirational districts (MoHFW, 2021).

Oxygen infrastructure must be strengthened to prevent future shortages. This includes functional PSA oxygen plants in all district hospitals, expanded liquid medical oxygen storage, oxygen pipelines in hospitals with more than 100 beds, and state-level oxygen reserves sufficient for 15 days. During the second wave, thousands of deaths occurred despite adequate production capacity of 7,100 metric tonnes per day, due to distribution failures. While over 1,500 PSA plants have been installed under PM-CARES, continued maintenance and an additional ₹10,000 crore investment are required (MoHFW, 2021).

Medical equipment stockpiling is also essential. India should maintain national reserves of PPE for three months, additional ventilators, testing kits, and essential medicines. Import dependence of 67% for medical devices caused severe shortages early in the pandemic. Domestic manufacturing incentives through the PLI scheme, with an allocation of ₹3,420 crore, can reduce vulnerability (Government of India, 2021).

Dedicated infectious disease hospitals should be established in all cities with populations above one million, along with isolation wards in medical colleges and advanced laboratories in all states. This requires an estimated ₹25,000 crore over ten years (MoHFW, 2021).

Non-Structural Measures

Early warning systems must be strengthened through expanded genomic surveillance under INSACOG, wastewater surveillance for early outbreak detection, and real-time disease reporting by strengthening the IDSP. These systems enable timely identification of new variants (NITI Aayog, 2021).

India also requires a dedicated pandemic preparedness law to define emergency powers, surveillance authority, and resource mobilisation. The Epidemic Diseases Act, 1897 proved inadequate for modern pandemic management.

Social protection must be expanded for informal workers, who constitute nearly 90% of the workforce, through portable social security, universal health coverage, and income support during emergencies. The lockdown exposed the absence of safety nets for this group.

Community preparedness should include training ASHA workers in pandemic response, public health literacy campaigns, mental health first aid training, and regular district-level simulation drills. Additionally, strengthening digital infrastructure—universal internet access, telemedicine platforms, and digital health records—is essential, as only 24% of households currently have internet access (NITI Aayog, 2021).

Policy and Planning Recommendations

Recommendation 1: Strengthen Public Health Infrastructure and Increase Health Spending

One of the biggest lessons from COVID-19 is that India's public health system needs much stronger investment. At present, public health spending is only about 1.28% of GDP, which is very low for a country of India's size. This should be increased to at least 2.5% of GDP over the next five years, with a long-term goal of moving closer to the WHO recommendation of 5% of GDP. Higher spending should be used to increase hospital and ICU beds to meet basic international norms, strengthen primary health centres in rural areas, and build dedicated infectious disease hospitals in major cities. This is important because around 65% of India's population lives in rural areas, while nearly 70% of doctors work in urban regions. Along with infrastructure, the health workforce also needs expansion by increasing medical and nursing seats and giving incentives for doctors and nurses to serve in rural and remote areas.

This recommendation is directly linked to Priority 3 of the Sendai Framework, which focuses on investing in disaster risk reduction for resilience. Strong health systems are critical infrastructure during pandemics. It also aligns with Sustainable Development Goal 3, especially Target 3.8 on universal health coverage, which stresses access to quality healthcare without financial hardship.

Implementation should include a separate budget line for health emergency infrastructure, use of public private partnerships in urban areas, greater central funding for aspirational districts, and regular independent monitoring of health system improvements.

Recommendation 2: Create a Dedicated Pandemic Preparedness Law

India also needs a clear and modern legal framework to deal with future pandemics. During COVID-19, authorities relied heavily on the Epidemic Diseases Act of 1897, which is outdated and not designed for today's complex health emergencies. A new pandemic preparedness law should clearly define how and when health emergencies are declared, who has decision-making authority, and how Centre-State coordination will work. It should make disease surveillance and data sharing mandatory, support genomic monitoring of new variants and allow fast emergency procurement of medical supplies. The law should also protect healthcare workers, include compensation mechanisms for people affected by emergency measures and ensure that emergency powers are time-bound and subject to parliamentary oversight.

This recommendation connects with Sendai Framework Priority 1 (understanding disaster risk) and Priority 2 (strengthening disaster risk governance), which highlight the importance of strong institutions and legal systems for managing disasters effectively.

Together, these policy reforms would make India far better prepared for future health crises.

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